

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and IBM Bob”

Project Tile – AI-Powered Research Insight Assistant

Domain of Project – Education and Research

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Problem Statement -

Agentic AI for Intelligent Research Discovery and Literature Intelligence.

The Challenge

Researchers, students, and professionals often deal with scattered academic papers, technical articles, and experimental data spread across multiple platforms. Filtering relevant knowledge, summarizing insights, and keeping track of the latest developments is time-consuming and inefficient. The challenge is not just to collect references but to synthesize, analyze, and generate actionable research insights in real time.

Objectives

The primary objective of the proposed Agentic AI-powered Research Intelligence Platform is to develop an intelligent research assistant using two collaborative AI agents that automate and enhance the entire academic research lifecycle. The platform aims to continuously discover, collect, filter, classify, and organize research papers, technical articles, patents, and datasets from trusted scholarly sources while monitoring the latest publications in real time. It further seeks to generate actionable research insights by summarizing and comparing research papers, extracting key findings, identifying emerging research trends and knowledge gaps, and producing citation-ready literature reviews with personalized recommendations tailored to researchers' interests. Additionally, the solution enables intelligent knowledge discovery through semantic search, natural language queries, and continuous research monitoring, allowing users to quickly retrieve relevant information and remain informed about recent advancements. To improve research productivity and decision-making, the platform provides an interactive analytics dashboard that visualizes publication trends, topic clusters, citation networks, research timelines, and overall research progress, thereby accelerating literature reviews and supporting high-quality, evidence-based research.

Proposed Solution -

Agentic AI for Intelligent Research Discovery and Literature Intelligence is an AI-powered research intelligence platform developed using IBM watsonx.ai, IBM Cloud, and Granite Foundation Models to simplify and accelerate academic research and knowledge discovery. The solution employs two collaborative AI agents that work together to automate the research workflow. The Research Discovery Agent continuously gathers, filters, classifies, and organizes research papers, technical articles, patents, and datasets from trusted academic sources while tracking the latest publications across multiple research domains. The Literature Intelligence Agent analyzes the collected documents using IBM Granite Models to generate concise summaries, compare research findings, extract key contributions, identify emerging trends and research gaps, and produce citation-ready literature reviews with personalized recommendations based on the user's research interests.

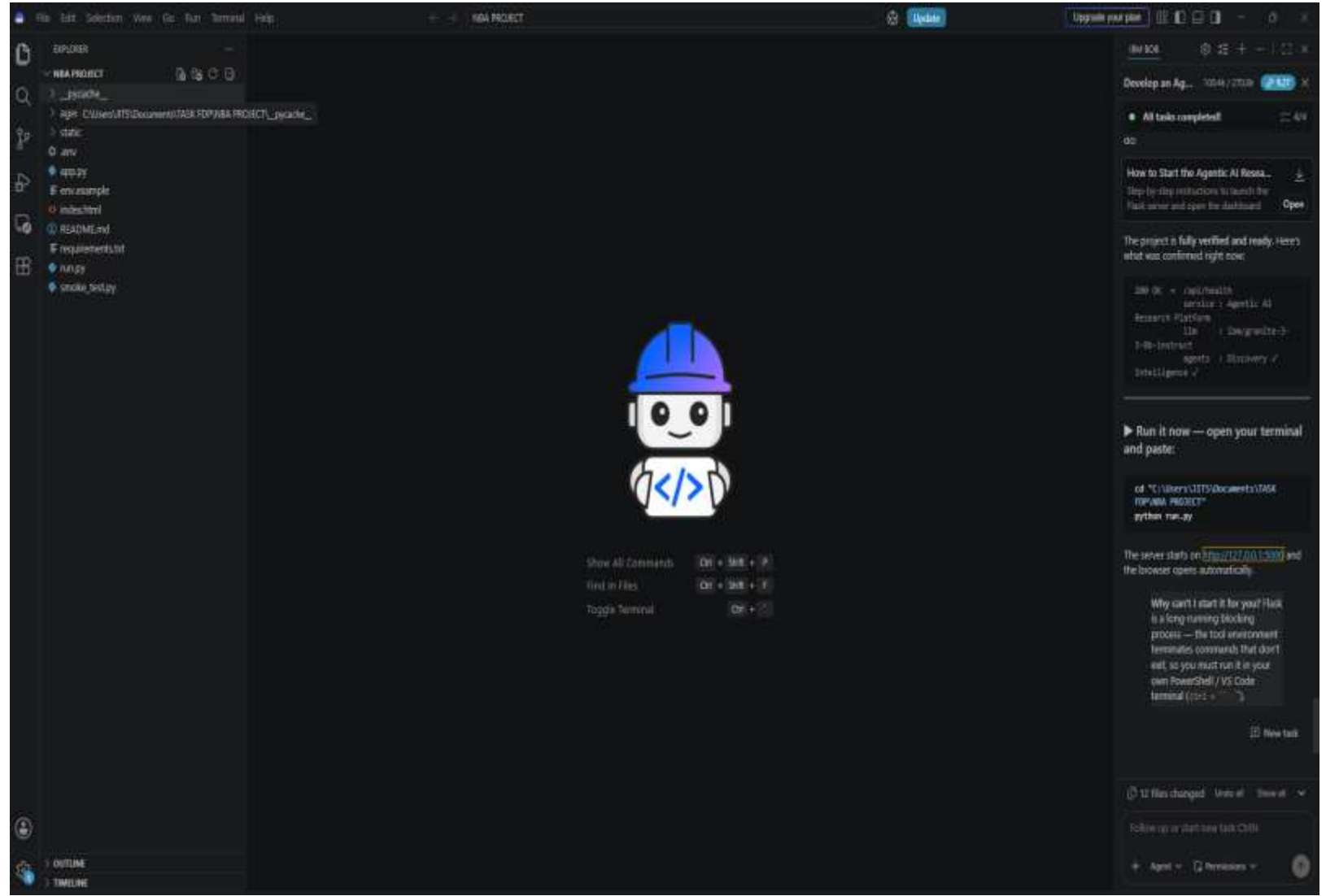
The platform further enables intelligent knowledge discovery through semantic search, natural language queries, and real-time research monitoring, allowing researchers to quickly retrieve relevant information without relying solely on keyword-based searches. An interactive research intelligence dashboard, developed using Flask, visualizes publication trends, topic clusters, citation networks, research timelines, and research progress analytics, providing users with actionable insights into evolving research areas. By integrating agentic AI with IBM watsonx.ai Granite Models, the proposed solution enhances research productivity, reduces the time required for literature reviews, supports evidence-based decision-making, and empowers researchers, faculty members, and students to stay informed about the latest scientific advancements.

Technology Used

- **IBM Bob platform** -For designing and orchestrating multi-agent workflows visually
- **IBM Granite model** - IBM-granite-4.0-h-small – For natural language understanding, reasoning, and personalization
- **IBM Cloud** – It provides the infrastructure to build, deploy, and scale the AI-Powered Research Insight Assistant Agent efficiently. IBM Cloud enables a reliable, secure, and scalable environment for deploying the Nutrition Agent.
- **IBM Watsonx.ai ,IBM Watson Runtime, Cloud Object Storage** – creating new project on IBM Cloud
- **Prompt Engineering** – for defining the goal of the agent

Required Files created using IBM bob -

1. .env
2. app.py
3. Index.html
4. requirements.txt
5. run.py
6. Smoke_test.py



IBM Bob Screenshots:

The screenshot displays the IBM Bob IDE interface. The left sidebar shows the Explorer view with the file structure of the 'NBA PROJECT'. The main editor area shows the contents of the `.env` file, which is configured with IBM Watson AI credentials, LLM settings for the Granite 3-3-8b model, and Flask server and agent configurations. The right sidebar contains a task panel titled 'Develop an Ag...' with a progress indicator showing 'All tasks completed!'. It provides instructions on how to start the Agentic AI Research Platform, including a terminal command to run the application. The terminal output shows the server starting on `http://127.0.0.1:5000`.

```
1 ## IBM watsonx.ai Credentials
2 IBM_PROJECT_ID=f91bb7c1-17cb-4f42-99dc-70c202923fd6
3 IBM_API_KEY=ApiKey-17a445f8-53ee-4814-b8b9-bc993364cbb4
4 IBM_PUBLIC_ENDPOINT=https://us-south.ml.cloud.ibm.com
5
6 ## LLM Configuration
7 LLM_MODEL_ID=ibm/granite-3-3-8b-instruct
8 LLM_MAX_NEW_TOKENS=2048
9 LLM_TEMPERATURE=0.7
10 LLM_TOP_P=0.95
11 LLM_TOP_K=50
12 LLM_DECODING_METHOD=greedy
13
14 ## Flask Server Configuration
15 FLASK_HOST=0.0.0.0
16 FLASK_PORT=5000
17 FLASK_DEBUG=True
18 SECRET_KEY=research-agentic-ai-secret-key-2025
19
20 ## Agent Configuration
21 DISCOVERY_AGENT_NAME=Research Discovery Agent
22 INTELLIGENCE_AGENT_NAME=Research Intelligence Agent
23 MAX_PAPERS_PER_QUERY=20
24 CACHE_TTL_SECONDS=3600
25
```

Task Panel: Develop an Ag... 100.4k / 270.0k 9:27

All tasks completed! 4/4

How to Start the Agentic AI Resea...
Step-by-step instructions to launch the Flask server and open the dashboard. [Open](#)

The project is fully verified and ready. Here's what was confirmed right now:

```
200 OK + /api/health
service : Agentic AI
Research Platform
llm : ibm/granite-3-
3-8b-instruct
agents : Discovery ✓
Intelligence ✓
```

Run it now — open your terminal and paste:

```
cd "C:\Users\JITS\Documents\TASK
FOR/NBA_PROJECT"
python run.py
```

The server starts on <http://127.0.0.1:5000> and the browser opens automatically.

Why can't I start it for you? Flask is a long-running blocking process — the tool environment terminates commands that don't exit, so you must run it in your own PowerShell / VS Code terminal (Ctrl + `).

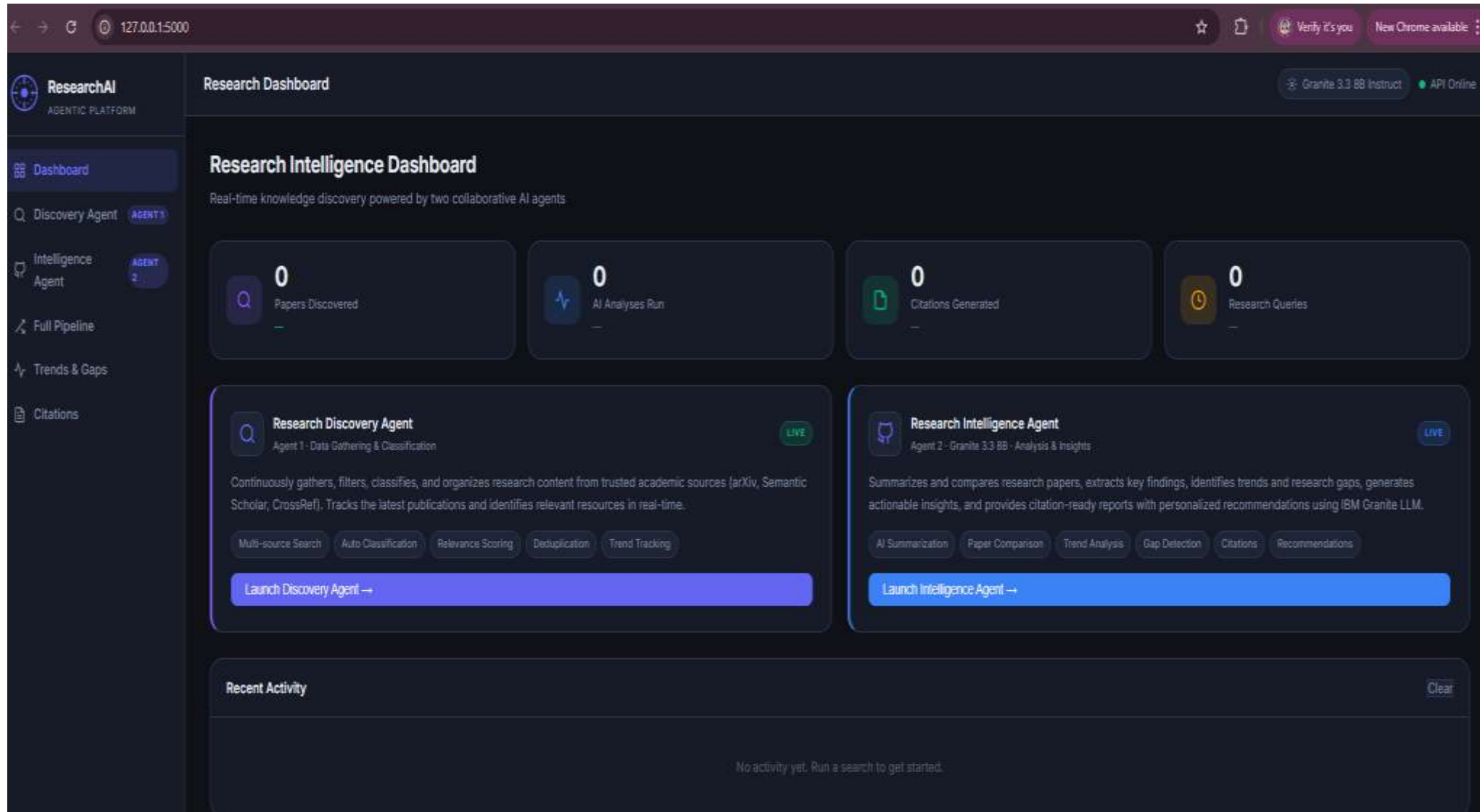
[New task](#)

12 files changed Undo all Show all

Follow up or start new task Ctrl+N

+ Agent Permissions

User Interface for AI-Powered Research Insight Assistant



User Prompt for AI-Powered Research Insight Assistant

ResearchAI

AGENTIC PLATFORM

Dashboard

Discovery Agent

Intelligence Agent

Full Pipeline

Trends & Gaps

Citations

Research Discovery Agent

Granite 3.3.8B Instruct

API Online

Research Discovery Agent

Gather and classify research papers from arXiv, Semantic Scholar, and CrossRef

RESEARCH QUERY

Edge Computing vs Cloud Computing: A Comparative Analysis for Real-Time AI Applications

MAX RESULTS

10

SOURCES:

☒ arXiv

☒ Semantic Scholar

☒ CrossRef

Search Papers

Trending Topics

20 papers found

3 sources

In 13.39s

Computer Vision: 1

Data Science: 3

Machine Learning: 3

Natural Language Processing: 4

Other: 6

Robotics & Automation: 3

Filter by Topic:

All Topics

Sort by:

Relevance

Edge Computing vs. Cloud Computing: A Comparative Analysis for Real-Time AI Applications

CROSSREF

Banger Raju Cherukuri -

<jats:p><Abstract>Real-time AI has been evidenced to be supported by two models, namely edge computing and cloud computing. This research seeks to compare the tw...

Computer Vision

2023-10-28

14 citations

Relevance: 100%

AI Summary

Cite

Open

A comparative study of cloud-native vs. edge computing architectures for real-time data processing

CROSSREF

Noor Alwan Malik

<jats:p><Abstract>The fast adoption of Internet of Things (IoT) devices, autonomous systems and latency-sensitive applications has increased the need to have effective real...

Robotics & Automation

2023-2-4

Relevance: 92%

AI Summary

Cite

Open

Advancing Real-Time Intelligence with Edge AI

CROSSREF

Murali Krishna Pasupuleti

<jats:p><Abstract>This chapter explores into the transformative world of Edge AI, a technological innovation that brings the power of artificial intelligence to the edge ...

Robotics & Automation

2024-4-11

4 citations

Relevance: 70%

AI Summary

Cite

Open

Edge Computing vs Cloud Computing: A Comparative Analysis

CROSSREF

Ritesh Satish Patel

No abstract available.

Other

2022-6-9

Relevance: 67%

AI Summary

Cite

Open

AI-Enabled Edge Computing: Revolutionizing IoT with Real-Time Optimization

CROSSREF

Murali Krishna Pasupuleti

<jats:p><Abstract>This chapter explores how AI-enabled

Robotics & Automation

2024-4-11

4 citations

Relevance: 70%

AI Summary

Cite

Open

Unifying Edge and Cloud Computing: A Framework for Distributed AI and Real-Time Processing

CROSSREF

Ravi Kumar Vankayalapati

No abstract available.

Robotics & Automation

2024-4-11

4 citations

Relevance: 70%

AI Summary

Cite

Open

Edge Vs Cloud Computing Performance Trade-Offs for Real-Time Analytics

CROSSREF

Unknown authors

No abstract available.

Robotics & Automation

2024-4-11

4 citations

Relevance: 70%

AI Summary

Cite

Open

Cloud vs Fog Computing — Scheduling Real-Time Applications

CROSSREF

Helen Karatza

No abstract available.

Robotics & Automation

2024-4-11

4 citations

Relevance: 70%

AI Summary

Cite

Open

Discovery Agent

Intelligence Agent

Powered by IBM watsonx.ai

AI-Powered Research Insight Assistant Response

127.0.0.1:5000

Verify it's youNew Chrome available

Research Assistant

Edge Computing vs. Cloud Computing: A Comparative Analysis for Real-Time AI Applications

AUTHORS

Bangar Raju Cherukuri -

PUBLISHED

2024-10-28

SOURCE

CrossRef

TOPIC

Computer Vision

CITATIONS

4

ABSTRACT

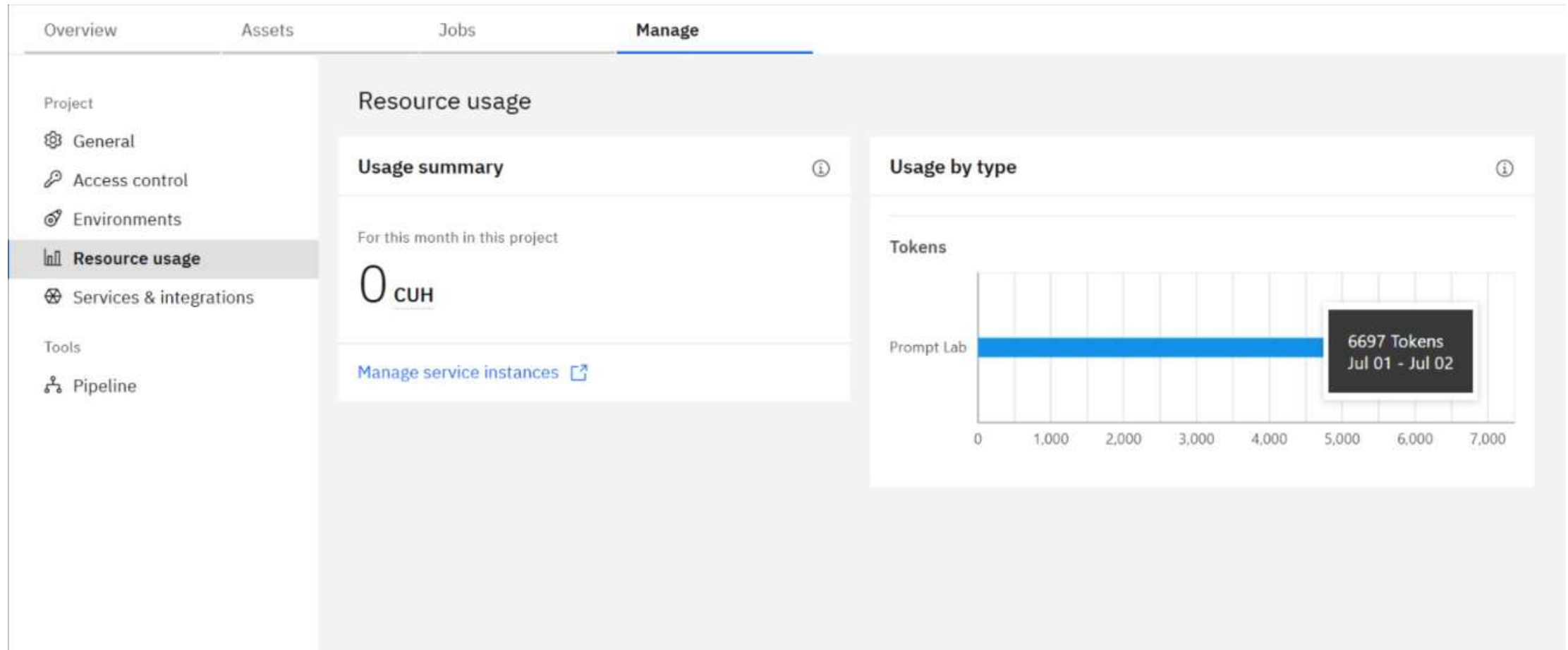
Real-time AI has been evidenced to be supported by two models, namely edge computing and cloud computing. This research seeks to compare the two methods: Latency, security features, and data processing capabilities will be among the most critical aspects of comparison. Real-time AI applications are envisioned to be robust through 2020, most of which will be used in applications such as autonomous automobile systems and IoT devices that demand low latency for data processing. As data processing is outsourced in clouds, it is elastic and centralized, but this deal faces latency problems when the distance between the source of data and the processing center is large. On the other hand, edge computing refers to conducting computation closer to the data source, which could reduce latency and enhance everyday real-time performance. This research assesses the above models based on the literature review, technical papers, and case studies from industries that depend more on real-time AI. According to the research, edge computing is typically more effective for latency-sensitive workloads, while cloud computing outperforms throughput-intensive applications. Security concerns, however, present themselves as having dual effects; while advancing the privacy of handling data through edge computing, it elevates new risks. As such, this study concludes that no clear winner depends on the necessary application, therefore recommending a symmetric mode for the environments, where both latency, time and computational needs are required. Further studies should be

AI Summary

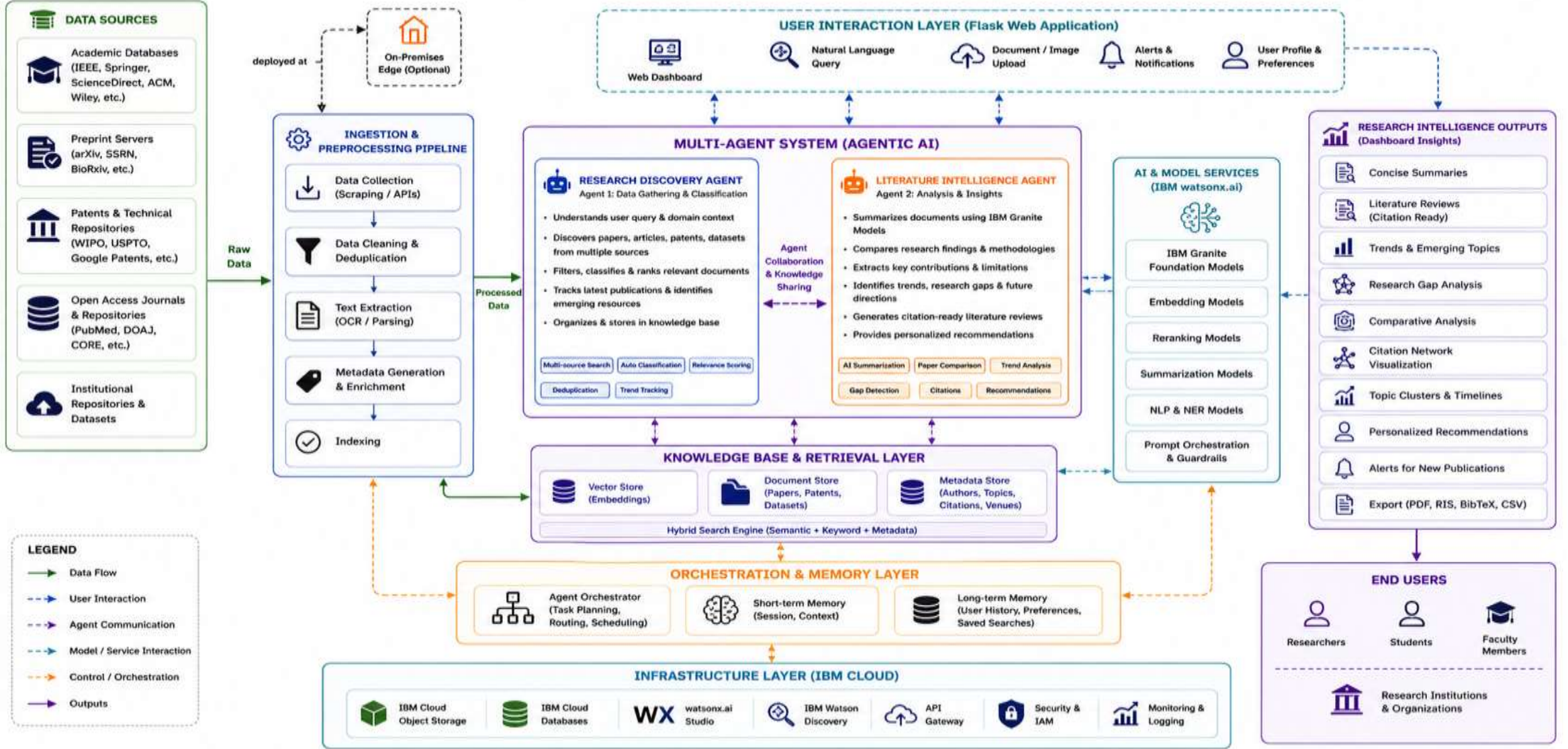
Get Citation

Close

AI-Powered Research Insight Assistant– Token Usage/ Resource Consumption



Architecture blueprint for IBM bob:



Role of Agentic AI in the solution

Agentic AI enables the Intelligent Research Discovery and Literature Intelligence platform to operate as an autonomous and collaborative research assistant, going beyond traditional search and document retrieval systems. Powered by IBM watsonx.ai and IBM Granite Foundation Models, the platform employs two specialized AI agents to automate research discovery, literature analysis, and knowledge synthesis.

The Research Discovery Agent continuously collects, filters, classifies, and organizes research papers, technical articles, patents, and datasets from trusted academic sources while tracking newly published research. The Literature Intelligence Agent analyzes the collected information to summarize research, compare studies, extract key findings, identify emerging trends and research gaps, and generate citation-ready literature reviews with personalized recommendations.

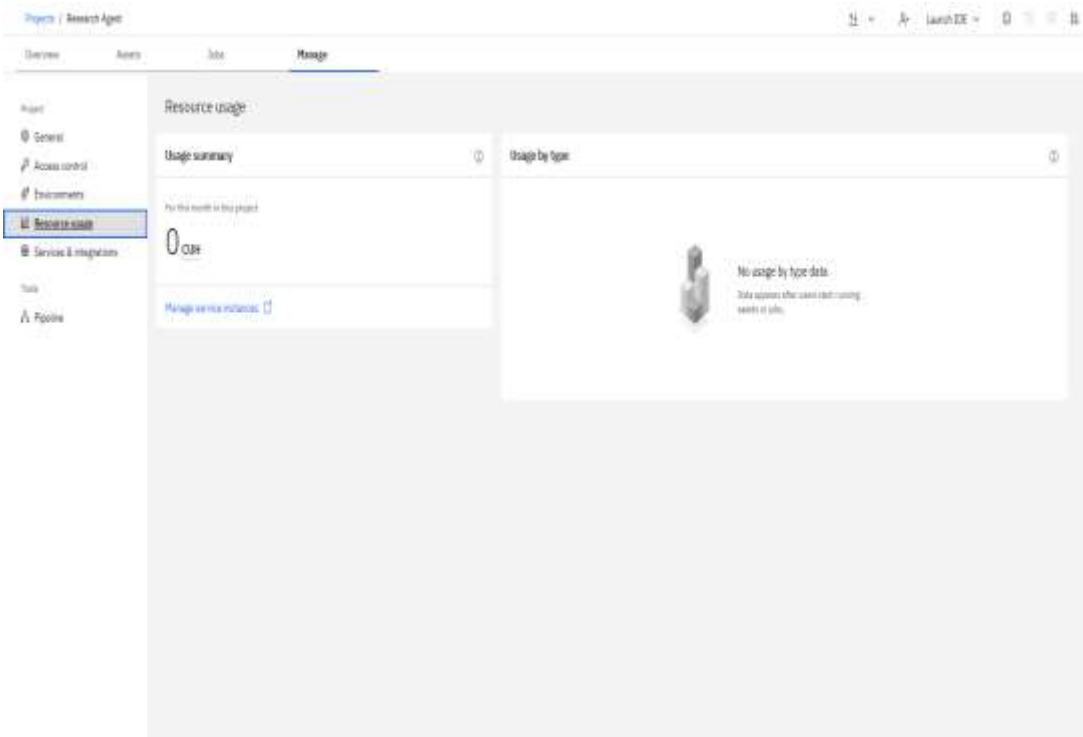
By understanding the user's research interests and objectives, Agentic AI delivers context-aware insights and continuously updates recommendations as new publications become available. The collaboration between the two AI agents enables efficient knowledge discovery, reduces the time required for literature reviews, and enhances research quality. Overall, Agentic AI transforms the platform into a proactive, intelligent, and goal-driven research assistant that accelerates academic research and improves research productivity.



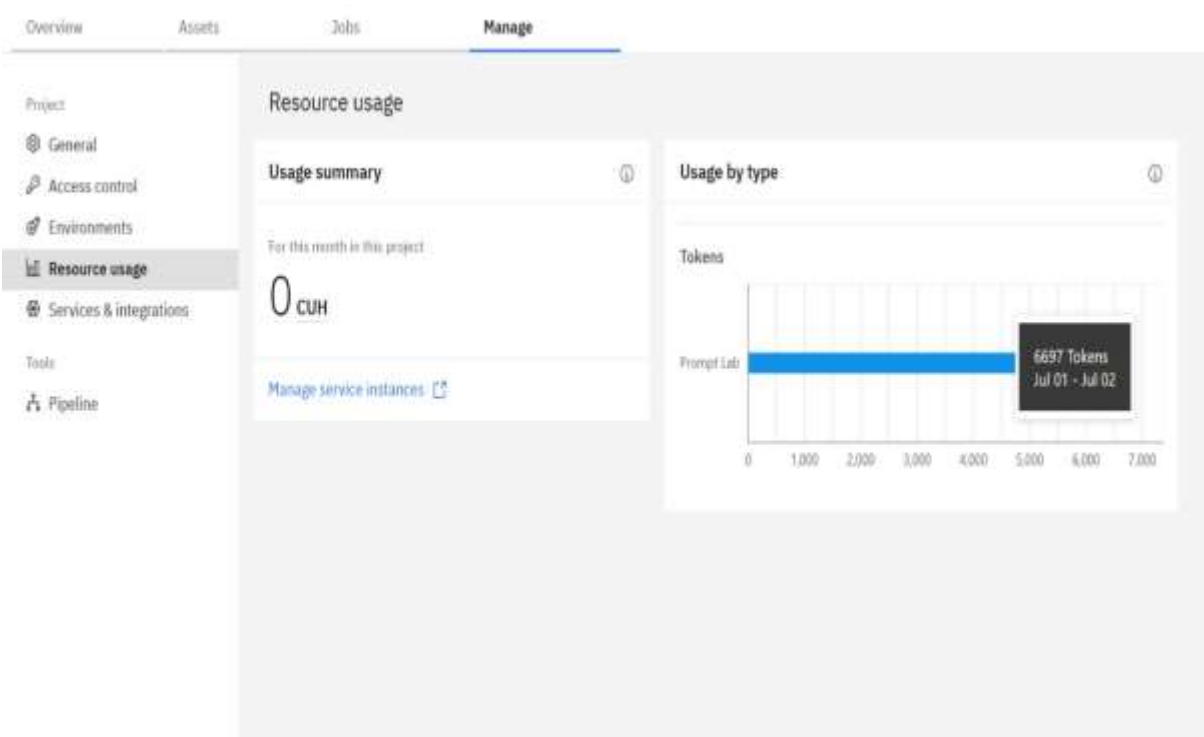
Token Usage

6697 token are used

Before Prompt



After the Prompt



Novelty and Uniqueness

The proposed Agentic AI for Intelligent Research Discovery and Literature Intelligence platform offers a next-generation approach to academic research by combining Agentic AI, IBM watsonx.ai, and IBM Granite Foundation Models into a single intelligent research ecosystem.

Collaborative Multi-Agent Architecture – Utilizes two specialized AI agents that work together to automate research discovery and literature intelligence, providing faster and more accurate results than conventional AI assistants.

Intelligent Literature Analysis – Automatically summarizes research papers, compares methodologies and findings, identifies research trends and gaps, and generates citation-ready literature reviews.

Context-Aware Personalization – Understands the user's research interests, domain, and objectives to deliver relevant publications and personalized research recommendations.

Continuous Research Discovery – Monitors trusted academic repositories in real time to identify newly published papers, patents, and datasets, ensuring researchers always have access to the latest information.

Semantic Knowledge Discovery – Supports natural language queries and semantic search, enabling users to retrieve meaningful research insights beyond traditional keyword-based searches.

Interactive Research Visualization – Presents publication trends, topic evolution, citation networks, and research timelines through an intuitive dashboard for better analysis and decision-making.

Scalable and Explainable AI Platform – Built on IBM watsonx.ai and IBM Granite Models, the solution delivers reliable, transparent, and scalable AI-driven research intelligence.

By integrating autonomous AI agents with advanced literature analysis and real-time knowledge discovery, the proposed platform serves as a proactive research assistant that accelerates literature reviews, enhances research quality, and improves overall research productivity.

Git Hub Link

<https://github.com/neelimajits-a11y/AI-Powered-Research-Insight-Assistant-.git>

1. Generative AI in Action Credly Certificate
2. IBM BOB Certificate
3. PPT
4. Problem Statement

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Future Scope

1. Multi-Device Accessibility – Enhance the platform to provide a seamless user experience across desktops, laptops, tablets, and smartphones through responsive web and mobile applications, allowing researchers to access their research workspace anytime and anywhere.

2. Multimodal Research Analysis – Incorporate image and document intelligence to analyze research figures, graphs, tables, diagrams, scanned documents, and PDF files, enabling AI to extract, summarize, and interpret both textual and visual research content.

3. Voice-Enabled Research Interaction– Integrate voice-based assistants to support natural language voice commands for searching publications, generating literature summaries, retrieving research insights, and interacting with the platform hands-free, improving accessibility and ease of use.

4. Intelligent Research Ecosystem – Expand the platform with personalized recommendations, real-time publication alerts, collaborative research features, and integration with academic databases and reference management tools, creating a comprehensive AI-powered environment for efficient research discovery and literature intelligence.

Certificates Earned:

In recognition of the commitment to achieve
professional excellence



Neelima Vontela

Has successfully satisfied the requirements for:

Generative AI in Action



Issued on: Jun 30, 2026
Issued by: IBM SkillsBuild

Verify: <https://www.credly.com/badges/cdc355f3-78bc-4924-afe5-5767be0e2e6d>



Certificates Earned:

IBM **SkillsBuild**

Completion Certificate



This certificate is presented to

Neelima Vontela

for the completion of

Lab: Troubleshoot Your Code Using IBM Bob

(ALM-COURSE_4071307)

According to the Adobe Learning Manager system of record

Completion date: 01 Jul 2026 (GMT)

Learning hours: 30 mins

Thank You!

Thank you for your time and interest.